

Lecture 4 Tree(I)

Rooted Trees

Basic Definition and Terminologies of Tree

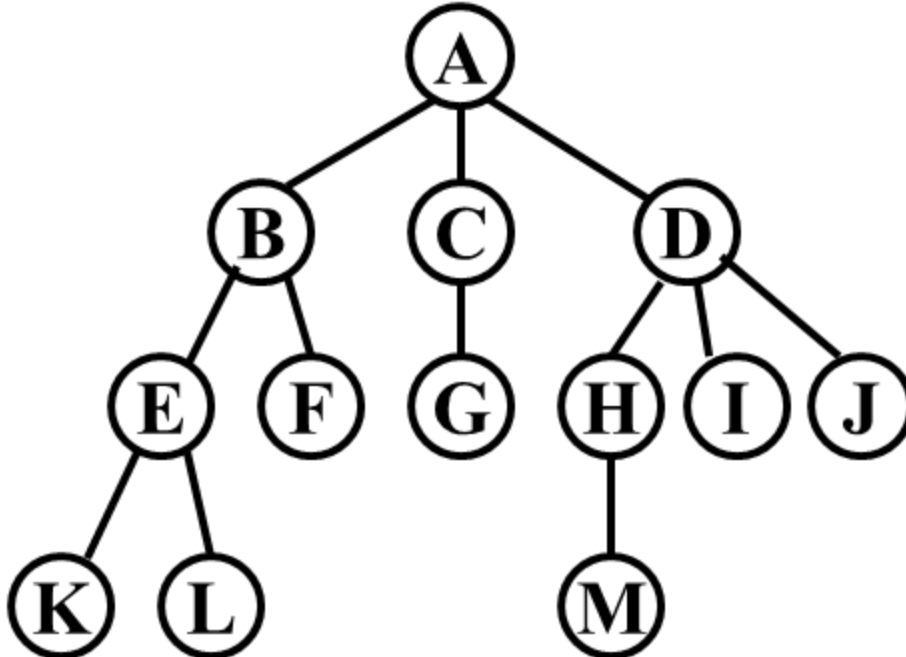
Omit.

Note that in PPT and textbook

- the level starts from 0 and root node is on the level 0
- Depth of a root node is 0
- Height of a leaf is 0.
- Height of an empty tree is -1.

Notation

- Graph Notation
- String notation:
The root comes first, followed by a list of subtrees.



(A)

(A (B, C, D))

(A (B (E, F), C (G), D (H, I, J)))

(A (B (E (K, L), F), C (G), D (H (M), I, J)))

Implementation

First child-Next sibling Representation:

Every node has data field and two pointer fields. One points to its first child and another point to the next sibling.

The representation is **not unique** since children can be any order.

Binary Trees

Basic Definition and Terminologies of Binary Tree

Omit.

Full binary tree:

A binary tree contains maximum of nodes in all levels.

Complete binary tree:

A binary tree is complete filled in each level except possibly the last, and all nodes are as far left as possible.

Maximum Number of Nodes in BT

- Maximum number of nodes on level l is $2^l, l \geq 0$.
- Maximum number of nodes of depth k is $2^{k+1} - 1, k \geq 0$.
- The number of nodes in a full binary tree of depth k is $2^{k+1} - 1, k \geq 0$.
- Complete binary tree of depth k has $2^k \leq n \leq 2^{k+1} - 1$ nodes.

Relations between Number of Leaf Nodes and Nodes of Degree 2

For nonempty binary trees, n_0 is the number of leaves, n_2 is the number of nodes of degree 2, then $n_0 = n_2 + 1$.

Implementation of Binary Trees

Array Representation

The nodes of the tree are sequentially in an array level by level from left to right for each level.

- The root node is indexed with 1.
- For any node indexed i , $1 \leq i \leq n$,
 - the left child of i is $2i$. ($2i \leq n$)
 - the right child of i is $2i + 1$. ($2i + 1 \leq n$)
 - the parent of i is $\lfloor i/2 \rfloor$. ($i \neq 1$)

Usually for complete binary tree.

Link Representation

```
typedef struct Treenode{
    ElemType data;
    TNode* left, right;
}TNode;
```

Binary Tree Traversal

- Preorder traversal
- Inorder traversal
- Postorder traversal
- Levelorder traversal

For expression trees, different traversal will produce corresponding ordered expression.